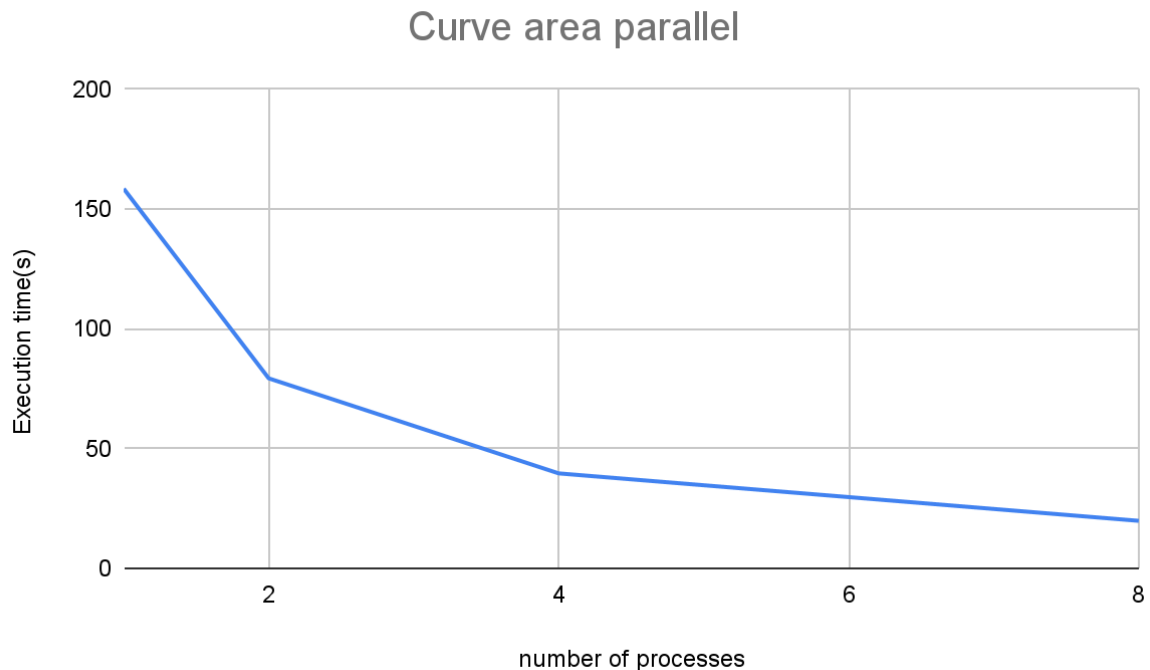


CMPT431 Assignment 6 Report Siu Yu Chau 301604828

Q1. Run your curve_area_parallel program from part 1 with 1, 2, 4, and 8 processes and the following parameters: coeffA=1.2, coeffB=0.8, rSeed=129, nPoints=4,000,000,000 (i.e., 4 billion). Each of your parallel programs should run 3 times. [Total number of runs is 4 (different process counts) x 3 (number of runs for each process count) = 12 runs]

Plot a graph with average execution time on the y-axis, process count on the x-axis.



Q2. From the plot in Q1, what is the parallel speedup for 2, 4, and 8 processes (compared to 1 process)? Is this problem embarrassingly parallel?

$$T_1 = 158.5, T_2 = 79.3, T_4 = 39.6, T_8 = 19.8$$

$$\text{Speed up for 2 processes} = 158.5 / 79.3 = 2.00$$

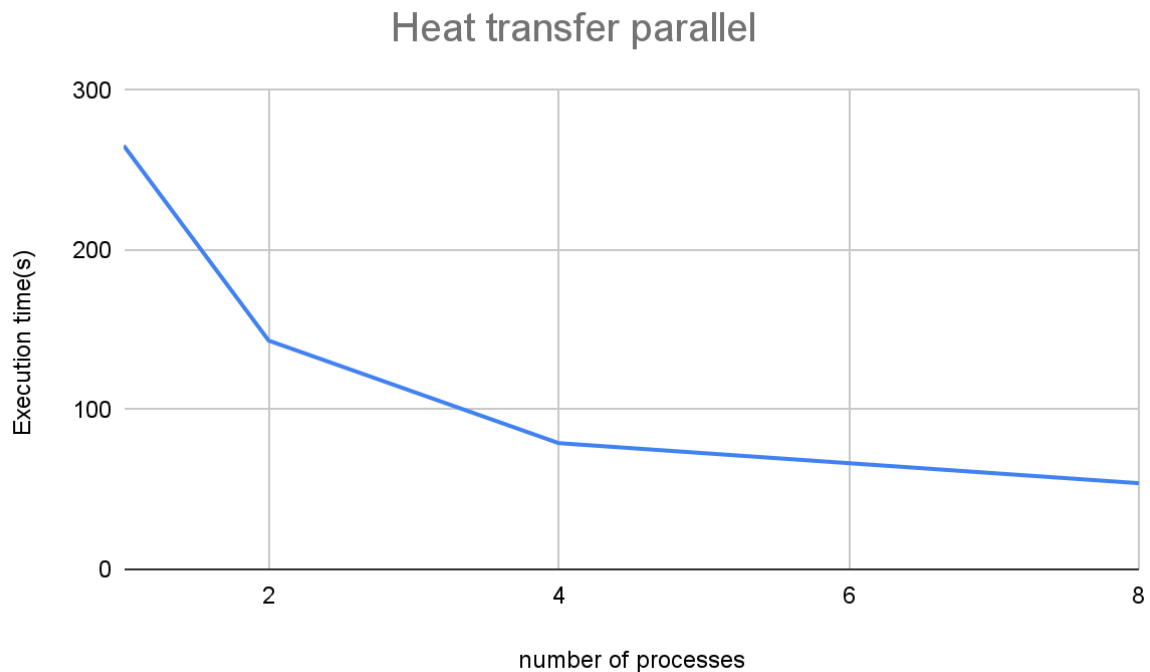
$$\text{Speed up for 4 processes} = 158.5 / 39.6 = 4.00$$

$$\text{Speed up for 8 processes} = 158.5 / 19.8 = 8.01$$

Yes this is an embarrassingly parallel problem because doubling the number of processes almost halved the execution time consistently.

Q3. Run your heat_transfer_parallel program from part 2 with 1, 2, 4, and 8 processes and the following parameters: gSize=4000, mTemp=600, iCX=0.15, iCY=0.1, tSteps = 1000 (if you already used 2000, that is also acceptable). Each of your parallel programs should run 3 times. [Total number of runs is 4 (different process counts) x 3 (number of runs for each process count) = 12 runs]

Plot a graph with average execution time on the y-axis, process count on the x-axis.



Q4. From the plot in Q3, what is the parallel speedup for 2, 4, and 8 processes (compared to 1 process)? Is this problem embarrassingly parallel?

$$T_1 = 265.4, T_2 = 143.2, T_4 = 79.0, T_8 = 53.9$$

$$\text{Speed up for 2 processes} = 265.4 / 143.2 = 1.85$$

$$\text{Speed up for 4 processes} = 265.4 / 79.0 = 3.35$$

$$\text{Speed up for 8 processes} = 265.4 / 53.9 = 4.92$$

No this is not an embarrassingly parallel problem because doubling the number of processes does not halve the execution time consistently. It has a diminishing return due to communication costs.